

Digital Transformation Strategies and Challenges in Small and Medium Enterprises (SMEs): A Systematic Review and Future Directions

Abstract

Small and medium enterprises (SMEs) are the backbone of the global economy, yet many struggle to harness digital transformation for competitive advantage. This paper provides a systematic review of recent literature (2020–2025) on digital transformation in SMEs, focusing on strategies employed, key technologies adopted, success factors, and challenges faced. Emphasis is placed on European SMEs, with comparative insights from the United States and Asia. The findings reveal that while digital transformation is increasingly crucial for SME productivity, innovation, and customer engagement, SMEs lag behind larger firms in adopting advanced technologies due to resource limitations, skill gaps, and other barriers. We identify common approaches SMEs use to navigate digital change – often incrementally and pragmatically – and highlight critical enablers such as leadership commitment, workforce upskilling, and supportive ecosystems. The discussion addresses theoretical implications for understanding SME innovation dynamics and offers practical recommendations for managers and policymakers to foster SME digitalization. Key gaps in current research are outlined, including the need for more cross-regional studies and longitudinal analyses. We conclude by proposing directions for future research to deepen knowledge on how SMEs can successfully transform in the digital era while overcoming inherent constraints.

Introduction

Digital transformation – the integration of digital technologies into business processes, models, and strategies – has become a priority across organizations worldwide. In recent years, its importance has escalated due to rapid technological advances and external shocks like the COVID-19 pandemic, which underscored the necessity of digital capabilities for business continuity and resilience ¹ ². SMEs, defined generally as firms with up to a few hundred employees, are particularly challenged and motivated in this context. On one hand, SMEs often operate with limited resources and digital expertise, making transformation difficult; on the other hand, digital tools can help level the playing field with larger competitors by enabling efficiency, innovation, and market expansion ³ ⁴.

SMEs hold a critical role in both the global and European economy (detailed in the next section), which makes their successful digitalization not only a firm-level concern but also a policy and development imperative. However, research indicates that SMEs have lagged behind large enterprises in adopting advanced technologies and reaping their benefits ⁵ ⁶. Many small businesses remain stuck with traditional processes, unable to exploit digital advantages due to knowledge gaps and resource constraints ⁷ ⁸. This paper aims to systematically review recent scholarly and institutional literature to explore how SMEs are approaching digital transformation, what challenges they encounter, and what outcomes and future pathways are emerging. The focus is on EU-based SMEs given the strategic emphasis on digitalization in Europe, and comparative insights from the USA and Asia are incorporated to broaden the

perspective. The study synthesizes definitions of digital transformation in the SME context, differences between SMEs and larger firms, key technologies (cloud computing, artificial intelligence, big data, Internet of Things) enabling transformation, and known success factors and barriers. We then suggest methodological approaches for researching this topic, present findings on common SME strategies and performance impacts, and discuss the implications. Finally, we conclude with recommendations and identify future research directions to support SME digital transformation moving forward.

Importance of SMEs in the Global and European Economy

SMEs are widely recognized as drivers of economic growth, employment, and innovation around the world. Globally, SMEs account for about 90% of all businesses and more than 50% of employment ⁹. In emerging economies, formal SMEs contribute up to 40% of national GDP on average (and even more if informal enterprises are included) ⁹. This economic significance is even more pronounced in Europe. Within the European Union (EU), SMEs (including micro, small, and medium firms) make up approximately 99.8% of all enterprises and employ about 64% of the workforce, contributing over half of the value added in the EU economy ¹⁰. In raw numbers, the EU counts over 24 million SMEs, which provided jobs for roughly 85 million people and generated about 51.8% of the EU's business sector GDP in recent analysis ¹⁰. Such enterprises span all sectors, from manufacturing and retail to services and tech startups, and are often hailed as the “backbone” of both national and regional economies.

Beyond Europe, similar patterns hold. In the United States, small businesses (broadly equivalent to SMEs) also comprise the vast majority of firms and around half of private-sector employment and GDP (a share that has slightly declined in recent decades, partly due to digital gaps) ¹¹. Across Asia, SMEs dominate national business landscapes – for example, Japan's economy is over 99% SMEs by firm count, and in developing Asian countries, SMEs are crucial for job creation and industrial development (though precise contributions vary by country). This ubiquity and importance of SMEs mean that their collective success or failure has significant implications for economic development, innovation diffusion, and social stability. Therefore, helping SMEs adapt to the modern digital economy is not just a micro-level concern but a macro-level priority.

The Increasing Relevance of Digital Transformation

As the global economy becomes more knowledge-based and technology-driven, digital transformation is increasingly viewed as essential for the long-term competitiveness and survival of businesses of all sizes. For SMEs, embracing digitalization offers a range of opportunities to improve performance, spur innovation, enhance productivity, and even compete on a more equal footing with larger firms ³. By adopting digital tools and platforms, SMEs can reduce operational costs and inefficiencies, gain richer business intelligence through data analytics, reach broader markets via e-commerce and online presence, automate and streamline processes, and differentiate their products or services through innovation. In short, digital technologies can allow a small firm to punch above its weight in terms of market reach and agility.

However, despite these significant benefits, many SMEs have been slow or cautious in their digital transformation. Smaller size often acts as a barrier to adoption – SMEs tend to have fewer financial and human resources, less in-house IT expertise, and less strategic capacity to plan and implement complex technological changes ¹². As a result, SMEs continue to lag in digital uptake compared to large

corporations, and this lag has consequences. Studies show that firms that fail to digitalize risk losing competitiveness: digital adopters typically invest more, innovate more, and grow faster than non-digital firms ⁶ . The COVID-19 pandemic highlighted this divide; during the 2020 crisis, many SMEs that lacked digital infrastructure struggled to maintain operations, whereas more digitally mature firms adapted more quickly (for example, by shifting to online sales or remote work) ¹ . The pandemic thus served as a catalyst in many regions, accelerating digital adoption out of necessity and raising awareness among SME owners about the strategic importance of going digital.

In Europe, digital transformation has been identified as a cornerstone of economic recovery and future resilience. The European Investment Bank's analysis noted that a sizable share of EU companies – especially smaller businesses – had not implemented any advanced digital technologies by 2020, and this slow pace *“threatens to impede...competitiveness in the long term”* ⁶ . Policymakers are therefore promoting initiatives to support SME digitalization, from funding high-speed internet infrastructure to providing training and grants for technology adoption ¹³ . Similar trends are observed in other parts of the world: for instance, governments in Asia (like Singapore's **SMEs Go Digital** program and Japan's subsidies for small manufacturers) are actively encouraging SMEs to modernize, seeing digital adoption as key to productivity and innovation gains. In summary, the relevance of digital transformation for SMEs is at an all-time high – it is increasingly regarded not as a luxury or IT matter, but as a strategic imperative for economic inclusion and business continuity in the digital age ² .

Literature Review

Definitions of Digital Transformation

The concept of “digital transformation” has been defined in various ways in recent literature, but common themes involve a fundamental change in business operations and value delivery enabled by digital technologies. One recent definition describes digital transformation *“as the process of doing things differently by creating new business designs through digital technology applications.”* This goes beyond mere digitization (automating existing processes) and entails modifying business models, altering value chains, and creating new products or services via digital means ¹⁴ . In other words, digital transformation is not just about adopting a specific IT tool, but about rethinking how an organization functions and delivers value in the digital era. It often involves a cultural shift and organizational change alongside technology implementation.

It is useful to distinguish **digitalization** from **digital transformation**. Digitalization typically refers to the process of converting analog processes to digital ones or using digital tools to improve efficiency (for example, moving from paper records to electronic databases). Digital transformation, in contrast, implies a more holistic integration of digital technology into all areas of a business, leading to significant changes in how the business operates and how it interacts with customers. This may include developing entirely new capabilities or revenue streams (such as an online platform for sales that did not exist before) and requires a strategic vision. For SMEs, digital transformation might manifest as adopting cloud-based software to automate accounting, leveraging e-commerce and social media to reach customers, or even redesigning the business model to offer digital products or services. The core idea is that digital transformation involves *“doing business differently”* in a way that leverages technology to drive substantial improvement or innovation in the firm's performance and value proposition ¹⁴ .

Digital Transformation in SMEs vs. Large Enterprises

SMEs differ markedly from large enterprises in their approach to and capacity for digital transformation. Large firms generally have more financial resources, IT staff, and formalized strategies, allowing them to invest early and heavily in cutting-edge technologies (such as advanced analytics, AI, or bespoke enterprise systems). SMEs, on the other hand, face resource constraints that make such investments more challenging ¹². Smaller companies often lack dedicated IT departments; many rely on external vendors or ad-hoc efforts by non-specialist staff. This leads to a digital adoption gap: for example, in the EU, less than 15% of SMEs (companies with 10+ employees, excluding large firms) have adopted artificial intelligence technologies, whereas over 41% of large enterprises have done so ¹⁵. Similar disparities are seen in the use of big data analytics, IoT, and other advanced tools – larger firms consistently outpace SMEs in these areas.

The literature highlights that *size* brings both disadvantages and some potential advantages for digital transformation. On the disadvantage side, SMEs suffer from economies of scale when implementing technology (the relative cost is higher for a small firm), have a harder time attracting tech talent, and often operate with short-term horizons that can make it difficult to pursue long-term digital strategies ⁸ ¹⁶. Organizational inertia and culture can also differ: while any company may resist change, in many SMEs the decision-making is concentrated in the hands of owners or a small management team, whose mindset (and sometimes lack of awareness) can either accelerate or stifle digital initiatives ¹⁷ ¹⁸. If the owner-manager does not possess a “digital mindset” or see the value of new technologies, the SME is unlikely to proactively invest in them.

On the advantage side, SMEs are often considered more agile and able to adapt quickly in certain situations. Their simpler structures can enable faster decision cycles and more flexibility in trying new things, compared to the bureaucracies of large corporations ¹⁹. This agility means that once SME leaders become aware of digital opportunities, they can implement changes with fewer layers of approval. Indeed, some studies note that smaller firms can leverage this agility to experiment with digital tools in niche areas or to customize solutions tightly to their market segment ²⁰ ⁴. In practice, however, many SMEs exhibit a *reactive* rather than proactive posture toward digital transformation: they tend to implement new technologies only in response to immediate pressures or customer demands, rather than as part of a long-term strategic plan ²¹. In contrast, large firms are more likely to have formal digital transformation roadmaps and dedicated budgets. Thus, a key challenge is how to enable SMEs to approach digital transformation more strategically despite their constraints – a theme that recurs in the research findings.

Key Technologies Driving SME Digital Transformation: Cloud Computing, AI, Big Data, and IoT

A range of emerging and mature technologies are commonly identified as the enablers of digital transformation for businesses. For SMEs in particular, the following technologies have been highlighted in recent literature and industry reports as key drivers or opportunities:

- **Cloud Computing:** Cloud technologies allow SMEs to use powerful software and IT infrastructure via the internet without large up-front investments in hardware or IT maintenance. Cloud-based applications (from storage and data backup to enterprise resource planning and customer relationship management systems) have lowered the barrier to entry for SMEs to implement sophisticated IT solutions ²². By paying subscription fees for software-as-a-service (SaaS) or

leveraging cloud platforms, SMEs can gain scalability and flexibility. Importantly, **cloud computing** offers cost-effective access to tools that were once affordable only for large firms. It also supports remote work and collaboration, which became especially crucial during the pandemic. The availability of cloud services means even micro-businesses can digitalize aspects of their operations (such as accounting, sales, or supply chain management) quickly and with minimal IT staff. Recent analyses emphasize that such service-based models enable SMEs to adopt technologies “without the need to purchase and manage the underlying technologies themselves,” thus reducing complexity and cost ²³ .

- **Big Data and Analytics:** The ability to collect, analyze, and draw insights from data (often termed **big data analytics**) is a cornerstone of digital transformation. For SMEs, big data may involve analyzing customer data, website traffic, sensor data from equipment, or market trends to inform decision-making. Using data analytics, even a small firm can identify patterns in customer behavior, personalize marketing, improve internal processes, or discover new market opportunities. In practice, however, many SMEs struggle to fully utilize data-driven approaches. Surveys indicate that a majority of SMEs find it challenging to effectively use the data they collect – for instance, 64% of SMEs reported difficulty in extracting actionable insights from their systems, and 74% felt they are not maximizing the value of their data investments ²⁴ . Challenges include a lack of skilled data analysts on staff and insufficient knowledge of analytical tools. Nonetheless, cases exist where SMEs have leveraged analytics successfully, leading to better customer targeting, inventory optimization, or product innovation. As data becomes increasingly available (from e-commerce metrics, social media, IoT devices, etc.), developing analytics capabilities remains both a key opportunity and a hurdle for SME digital transformation.
- **Artificial Intelligence (AI):** AI technologies, ranging from machine learning algorithms to natural language processing and computer vision, offer powerful means to automate and enhance business tasks. SMEs can potentially use AI for functions like chatbots in customer service, predictive analytics for sales forecasting, or intelligent process automation in operations. The transformative potential of AI is significant – studies in Europe have found that AI adoption by SMEs can positively impact revenue growth (one study noted an average revenue increase of over 30% associated with AI use) ²⁵ . However, adoption of AI among SMEs is currently quite limited. As noted earlier, only about 13% of EU enterprises with 10+ employees were using any AI technology as of 2024, compared to 41% of large firms ¹⁵ . Barriers include the complexity of AI solutions, high implementation costs for advanced AI, and a pronounced skills gap – SMEs may not have data scientists or AI experts to deploy these systems. There are also concerns about understanding AI use-cases and trustworthiness. Even so, accessible forms of AI (such as AI features built into cloud software, or third-party AI services) are gradually making their way into SME operations. For example, small retailers might use AI-driven analytics in their online advertising platforms without developing the algorithms in-house. Over time, as AI tools become more user-friendly, SMEs are expected to increase adoption, especially in areas that can directly improve efficiency or customer engagement (like automated customer support, personalization, or demand forecasting).
- **Internet of Things (IoT):** The IoT refers to networks of physical devices (sensors, machines, vehicles, appliances, etc.) connected to the internet, which can collect and exchange data. In an industrial or commercial context, IoT enables real-time monitoring of processes, predictive maintenance of equipment, supply chain tracking, and smarter inventory management, among other applications. For manufacturing SMEs or those in sectors like logistics and agriculture, IoT can be a game-changer

by providing visibility and control that improve productivity and reduce downtime. For instance, a small manufacturing firm can use IoT sensors on machines to predict failures and schedule maintenance only when needed, thereby saving costs. Despite these benefits, integrating IoT is often challenging for SMEs. One issue is that many SMEs rely on older, legacy equipment not designed for connectivity – **upgrading to IoT-enabled machinery can require significant capital investment**. It has been estimated that achieving an Industry 4.0 level of smart manufacturing could require replacing or retrofitting 40–50% of existing machinery to accommodate IoT sensors and related software ²⁶. Such an undertaking is daunting for a smaller firm. Moreover, IoT implementations generate large volumes of data, necessitating data management and analysis capabilities (tying back to the analytics challenge). Security is another concern, as connecting devices can introduce vulnerabilities if not properly managed. Nonetheless, some SMEs have embraced IoT through incremental steps or by using third-party solutions. For example, logistics SMEs might adopt GPS trackers and simple sensor devices to monitor fleet and cargo conditions. As costs of IoT hardware decrease and platform solutions (IoT-as-a-service) grow, more SMEs are likely to experiment in this area, especially if they operate in production-centric industries.

In summary, cloud computing, big data/analytics, AI, and IoT stand out as four technological pillars frequently discussed in relation to SME digital transformation. They often intersect with each other (e.g., cloud-based AI analytics platforms, or IoT devices streaming data to cloud databases for analysis). It is important to note that SMEs do not adopt technology for technology's sake – the adoption is usually motivated by specific business needs such as improving customer reach, reducing costs, or solving operational bottlenecks. Simpler digital tools like having a mobile-responsive website, using social media for marketing, or adopting electronic payment systems can also be part of an SME's digital transformation journey, even if they are not “advanced technology” per se. In fact, studies have observed that many SMEs begin digitalization with relatively basic ICT usage (email, office software, social media), then move into more sophisticated areas once they build confidence and see returns ²² ²⁷. The challenge and opportunity for SMEs lie in leveraging these technologies appropriately to innovate and remain competitive, within the confines of their limited resources.

Success Factors and Barriers in SME Digital Transformation

A rich body of recent literature has analyzed what factors lead to successful digital transformation in SMEs and what barriers frequently impede progress. Understanding these success factors and obstacles is crucial for developing effective strategies and support mechanisms for SME digitalization.

Success Factors: Several key enablers or critical success factors for SME digital transformation emerge repeatedly in studies:

- **Strong Leadership and Vision:** Top management support is often cited as a decisive factor. When SME owners or managers have a clear vision for digital transformation and actively champion it, projects are more likely to succeed ²⁸ ²⁹. Leadership involves setting a digital strategy aligned with the company's goals and allocating sufficient resources to achieve it. In small firms, the leadership's digital mindset can make or break transformation initiatives.
- **Alignment with Business Strategy and Processes:** Successful SMEs treat digital transformation as an integral part of their overall business strategy, not a standalone IT project ²⁹. This means identifying how specific technologies can solve business problems or create value, and then

redesigning processes accordingly. SMEs that carefully evaluate their *baseline* (current capabilities, resources) and tailor their digital roadmap to their unique context tend to implement more effective solutions ³⁰. A one-size-fits-all approach does not work; the transformation must fit the firm's operational realities and market environment.

- **Incremental Implementation and Agility:** Given resource constraints, SMEs often benefit from an incremental approach to digital transformation ³⁰. Rather than attempting a massive overhaul in one go (which could be risky and costly), successful SMEs start with small projects or pilot implementations, learn from them, and gradually scale up the digital innovations. This iterative approach allows for flexibility and learning, reducing the risk of failure. It also helps in managing change within the organization, as employees can adapt step by step. Agility – the ability to respond quickly to feedback and changing conditions – is a natural advantage of many SMEs and serves them well in the transformation process ¹⁹.
- **Skilled and Adaptable Workforce:** Human capital is a critical element. SMEs that invest in developing their employees' digital skills and knowledge create a foundation for successful adoption of new technologies ¹⁷ ³¹. Training programs, hiring tech-savvy staff, or partnering with external experts are ways to enhance capabilities. A digitally literate workforce can better understand the benefits of new systems, reducing resistance to change and increasing effective use of tools. In fact, a lack of digital skills is identified as one of the biggest barriers (as discussed below), so overcoming that by fostering continuous learning and skill-building is a success strategy ³⁰. Some SMEs cultivate a culture of innovation and learning, encouraging employees to experiment with new ideas and technologies, which supports transformation.
- **Effective Change Management and Culture:** Beyond technical skills, the organizational culture must be supportive of innovation. SMEs that manage change well – by communicating the purpose of digital initiatives, getting buy-in from employees, and perhaps rethinking organizational structures – are more likely to see their projects bear fruit. Cultivating a culture open to new ideas and not afraid of failure is helpful. In small firms, where employees often wear multiple hats, ensuring that everyone understands the value of a new digital tool can significantly smooth the implementation.
- **External Support and Partnerships:** SMEs do not operate in isolation; leveraging external networks is a success factor noted in several studies. This can include partnerships with technology providers (who may offer training or customized solutions), collaboration with other firms (sharing best practices), and utilizing government or industry support programs ³² ³³. For example, joining a government-sponsored digital innovation initiative or working with a local university on a tech pilot can provide SMEs with expertise and resources they lack internally. An ecosystem of support – including consultants, suppliers, and even customers – can drive SME success in digital projects.
- **Financial Resources and Planning:** While SMEs have limited budgets, those that allocate funds wisely and seek external financing when needed (e.g., grants, subsidies, loans for digital investment) are better positioned. A key success factor is conducting cost-benefit analysis to prioritize high-impact digital investments. Access to finance remains a constraint for many, but forward-looking SMEs plan for digital transformation in their budgeting and make a business case for it (even if ROI might be realized in the longer term). The literature suggests that coupling financial planning with a

clear strategy (as mentioned above) ensures that resources – however limited – are focused on the most strategically relevant technologies ³⁴ .

Common Barriers: On the flip side, SMEs encounter a number of recurring barriers that slow or derail digital transformation efforts. Recent research (including systematic reviews and large-scale surveys) have pinpointed the following major obstacles:

- **Insufficient Financial Resources and High Cost of Investment:** Limited budget is an almost universal challenge. Many SMEs find the upfront costs of new technologies (software licenses, new equipment, IT infrastructure upgrades) to be prohibitive ³⁵ . Unlike large firms, they cannot as easily absorb the cost of experimentation or failures. The expense of not just acquiring technology but also maintaining it and training staff can deter investment. High perceived cost relative to uncertain benefit is a barrier, especially for cutting-edge technologies like AI or IoT which may not have immediate payback. This financial constraint is often cited as the top barrier in surveys of SME digitalization ³⁵ .
- **Lack of Digital Skills and Expertise:** A critical internal barrier is the shortage of skills – both technical IT skills and broader digital literacy – within SME workforces ¹⁷ ³⁵ . SMEs often do not have IT specialists, and existing staff may not have training in areas like data analytics, cybersecurity, or software integration. This skills gap means SMEs may not even know where to start or how to implement digital tools effectively. One study called the lack of digital skills (and mindset) *“one of the major barriers for SMEs”*, noting that without the necessary expertise, small firms fail to recognize the benefits or understand how to integrate new technologies, breeding skepticism and resistance ¹⁷ . In essence, the human factor – not having the right knowledge – can halt transformation even if technology is available.
- **Resistance to Change and Cultural Barriers:** People naturally resist change, and in an SME, where employees have done things a certain way for years, introducing new digital processes can meet with pushback. Workers may fear job loss (e.g., due to automation) or simply be reluctant to learn new systems. Additionally, if leadership is not fully convinced, their ambivalence can translate into organizational inertia. A *“business-as-usual”* culture that does not prioritize innovation will struggle to embark on digital transformation ²⁸ ³⁶ . Some research specifically highlights older or family-run SMEs where traditional mindsets dominate, making them slower to adopt new technologies (preferring proven methods). Overcoming this requires change management, as noted under success factors, but until then it stands as a barrier.
- **Lack of Strategic Planning and Vision:** Many SMEs undertake digital adoption in a piecemeal or reactive way without an overarching strategy. The absence of a clear digital transformation roadmap can lead to misaligned projects, wasted resources, or stalled initiatives. For example, an SME might implement an e-commerce solution but fail to integrate it with inventory management, resulting in operational inefficiencies. Several authors point out that a *“long-term vision”* is often lacking – SMEs may not think about data management, scalability, or cybersecurity until problems arise ³⁷ . This short-term, ad-hoc approach is a barrier because digital transformation typically requires coherent planning across the business. The literature suggests that many SMEs need guidance in developing digital strategies that map technology investments to business goals ²⁹ ³⁷ .

- **Inadequate Infrastructure and Technological Issues:** Some SMEs, especially in certain regions, face infrastructural barriers such as unreliable internet connectivity, lack of access to advanced technologies, or legacy systems that are incompatible with new applications. In developing areas or rural regions, an SME might want to digitalize but doesn't have access to high-speed broadband, for instance. Even within advanced economies, a small firm might be running on outdated computers or software that make new implementations difficult. Integrating new digital tools with old systems can be complex and costly (e.g., linking an online store with an old inventory software). Additionally, concerns about IT security and data privacy constitute barriers – SMEs are wary of cyber risks and may feel they cannot adequately secure new digital systems, leading to hesitation in adoption ³⁵ ³⁷. High IT security requirements and regulations (like GDPR in Europe) can be daunting for a small firm to navigate without expert help.
- **Finding the Right Technology and Partners:** SMEs often lack information about what solutions are available and suitable for their needs. A barrier noted is *“lack of knowledge about technology providers and options”* ³⁵ – unlike large firms that might have consultants and R&D departments scanning the tech landscape, SMEs may be unsure which software or platform to choose, whom to trust as a vendor, or how to evaluate competing solutions. This can lead to analysis paralysis or choosing suboptimal tools. Relatedly, if an SME selects a technology but doesn't get proper support from the provider (or can't afford full support), the implementation might fail, reinforcing a fear of digital projects.
- **Regulatory and External Constraints:** While less commonly cited than the above factors, some SMEs face external barriers like complex regulatory compliance issues when going digital. For example, moving data to the cloud might raise compliance questions about data protection; adopting e-commerce could mean navigating consumer protection laws and tax regimes in multiple jurisdictions. SMEs have limited capacity to deal with such complexity. Also, in certain industries, heavy regulation can divert resources – one study on European family SMEs noted that **increasing regulatory requirements** (e.g., for data protection) consumed management attention and resources *“often at the expense of a digital transformation strategy”* ³⁸. Thus, the external environment can pose hurdles if not SME-friendly.

In summary, the success of SME digital transformation hinges on leveraging enablers (leadership, skills, strategy, incremental approach, external help) while mitigating the many barriers (lack of funds, skills, cultural resistance, etc.). The interplay of these factors determines whether an SME merely adopts a few digital tools or truly transforms its operations. The literature consistently underscores that overcoming barriers like skill gaps and financial constraints will require concerted efforts not just by the SMEs themselves but by a wider ecosystem, including policymakers, educators, and technology providers ³⁹ ³³. This is because many obstacles (e.g., limited access to capital or training) cannot be solved by an individual small firm alone. Thus, understanding these factors is crucial for designing interventions and strategies, a point we return to in the Discussion.

Methodology Suggestions (Research Design for Studying SME Digital Transformation)

Conducting research on digital transformation in SMEs can be approached through various methodologies, each offering different insights. Given that our paper is a systematic review of existing literature, we summarize prevailing methods and suggest suitable approaches for primary research in this domain:

Systematic Literature Review: First, to gather the current state of knowledge (as done in this study), a systematic literature review (SLR) methodology is appropriate. Following best practices (such as PRISMA guidelines), researchers can define inclusion criteria (e.g. peer-reviewed articles from 2020–2025 on SMEs and digital transformation), search multiple databases, and then synthesize findings from the selected studies. An SLR helps in identifying common themes, research gaps, and consensus or divergences in the literature ⁴⁰ ⁴¹. In our review, for instance, we used thematic analysis on the collected studies to extract core success factors and strategies for SMEs ⁴¹. An SLR is a robust method for understanding what has been studied and where further investigation is needed, forming a foundation for primary research.

Quantitative Surveys: For primary empirical research, one common approach is to conduct surveys of SMEs to collect quantitative data on their digital transformation status, practices, and outcomes. Surveys can reach a broad sample of firms, allowing for generalizable findings about trends and correlations. Researchers might design a questionnaire asking SME owners/managers about things like which technologies they have adopted, perceived barriers (using Likert-scale measures), and any performance changes observed. With a sufficient sample size, statistical analysis can be applied. Techniques such as regression analysis or structural equation modeling (SEM) are often used to test hypotheses (for example, examining the relationship between digital adoption and firm performance or the influence of organizational factors on transformation success). Indeed, recent studies note that survey research with SEM is a standard quantitative method in this field, useful for testing conceptual models of digital transformation drivers and outcomes. For instance, a study aiming to build a framework for SME digital resilience used a survey-based approach combined with SEM to validate the relationships between strategy, technology adoption, and resilience outcomes ⁴². When doing cross-regional comparisons (EU vs. USA vs. Asia), a standardized survey distributed in multiple countries could reveal differences in adoption rates and challenges under different environmental conditions.

Qualitative Case Studies: On the qualitative side, case study research is highly valuable for gaining deep, contextual insights into how individual SMEs plan and execute digital transformation. A case study could involve one or multiple SME cases. Researchers collect data through interviews (with owners, managers, employees), on-site observations, and documents (strategy documents, project reports, etc.). This method allows exploration of the *process* of transformation – for example, how did a particular manufacturing SME implement IoT and what challenges did it face along the way? What strategies did a retail SME use to train staff for e-commerce adoption? Case studies can illuminate the nuanced interplay of factors (like culture and leadership) that surveys might not fully capture. They are especially useful to build or refine theory, and to understand causal mechanisms in detail. For example, an exploratory study on SME family firms used a **multiple case study** of eight firms (with 19 interviews) to explore how overconfidence and centralized decision-making affected digital strategy, yielding rich insights into internal dynamics ⁴³. Similarly, case studies have been used to document *successful* digital transformation roadmaps, providing practical examples or “best practices” that others can learn from ⁴⁴. We suggest that future research could employ

longitudinal case studies, where an SME is followed over time through its digital transformation journey, to observe how changes unfold and what impacts occur at each stage.

Mixed-Methods Approaches: A powerful approach can be to combine quantitative and qualitative methods. For instance, a researcher might start with a broad survey of SMEs in a region to identify patterns (e.g., which barriers are most pervasive, or which technologies correlate with better performance), and then follow up with case studies or in-depth interviews with a subset of firms to understand the story behind the numbers. This mixed-methods design can provide both generalizability and depth. Another approach is to use **Delphi studies or focus groups** with experts and SME stakeholders to build consensus on future trends or necessary policy interventions – capturing insights that might not be evident from firm-level data alone.

Cross-country Comparative Studies: Given our focus on EU vs. USA vs. Asia comparisons, researchers should also consider comparative designs. This could entail harmonized surveys in multiple countries or comparative case studies. For example, one might compare SMEs in a high-tech Asian economy (like South Korea) with those in a European country (like Italy) to examine how cultural and policy differences influence digital transformation. Care should be taken to account for differing definitions of SME and economic contexts. Institutional reports often provide baseline data (e.g., OECD, Eurostat), which can guide sample selection and contextualization.

In conducting primary research, it is crucial to address the typical challenges: ensuring a representative sample of SMEs (which are a very heterogeneous group), obtaining honest and accurate information (SME leaders may over- or under-estimate their digital maturity), and dealing with rapidly changing technology (today's frontier tech might be mainstream tomorrow, so findings must be timely). Triangulating data from multiple sources (e.g., combining survey responses with some objective performance data, or interviews with some direct observations of digital tools in use) can improve reliability. Finally, ethical considerations such as data privacy should be respected, especially when SMEs share sensitive information about their operations.

In summary, a multi-method research strategy is suggested to comprehensively study SME digital transformation. Quantitative methods yield broad patterns and potential causal links, while qualitative methods reveal deep insights and mechanisms. By using these methods in complementary ways, researchers can build a robust understanding of how SMEs navigate the digital era and how various factors contribute to success or failure.

Findings

Common Strategies Used by SMEs in Digital Transformation

The literature indicates that SMEs often approach digital transformation in ways that reflect their resource constraints and need for practicality. Unlike large firms that might deploy a grand digital overhaul, SMEs tend to adopt more modest, stepwise strategies. Several common strategic patterns have been identified:

- **Alignment with Business Needs and Context:** Successful SMEs begin by assessing their own business context – their size, sector, market, and internal capabilities – and then develop a digital strategy that aligns with these factors rather than adopting technology for its own sake ³⁰. For example, a small B2B manufacturer might prioritize digital tools for supply chain and production

efficiency, whereas a consumer-facing retail SME might focus first on online sales channels and digital marketing. Recognizing one's *"baseline, limitations, and idiosyncrasies"* is crucial ³⁰. In practice, this means SMEs often start with a few targeted initiatives that address pressing business problems or opportunities (like setting up an e-commerce site if customers are demanding online purchasing, or implementing a cloud CRM to better manage leads if sales growth is a goal).

- **Incremental and Iterative Implementation:** Rather than big-bang transformations, SMEs typically use incremental steps. They roll out new technologies in a phased manner – for instance, piloting a new software in one department or for one product line, then expanding if successful ³⁰. This *"gradually improving digitalization"* approach allows learning by doing and reduces risk. Studies suggest that SMEs should indeed *"start with incremental"* changes and scale up over time ³⁰. Many SMEs implement a series of small projects (like digitizing payroll this quarter, launching social media marketing next quarter, then integrating an inventory management system) that cumulatively lead to substantial transformation. This incremental strategy aligns with agile principles and helps maintain operations without large disruptions.
- **Pragmatic Use of Off-the-Shelf Solutions:** Lacking the capacity to develop custom tech solutions, SMEs commonly rely on readily available digital tools and platforms. Cloud services, in particular, are a popular choice (as noted earlier) because they provide quick solutions that are relatively easy to adopt. SMEs often choose software that is affordable and user-friendly, even if not perfectly tailored, and then adapt their processes around it. For instance, an SME might use a generic accounting software or a standard e-commerce platform like Shopify or WooCommerce rather than developing a custom system. This pragmatism extends to using free or low-cost digital resources (such as leveraging social media for marketing or using freeware for project management) – essentially getting on the digital ladder with minimal cost. While such choices may not be optimized for scale, they allow SMEs to make tangible progress in digitalization with limited investment.
- **Focus on Customer-Facing Digitalization First:** Many SMEs prioritize digital initiatives that have immediate impact on sales and customer engagement. This is reflected in the high adoption of tools like social media marketing, online marketplaces, and digital payment systems among SMEs ²². Establishing an online presence (website, social media profiles) is often one of the first steps in SME digital transformation. It directly connects to revenue and is relatively straightforward. By improving customer-facing processes (e.g., online ordering, digital customer service via chatbots, email marketing), SMEs aim to enhance customer experience and expand market reach without needing extensive internal re-engineering. Once these front-end elements are in place, some SMEs then turn to back-end or internal process digitization (such as enterprise software for operations). This outside-in approach ensures that the digital strategy quickly supports business growth.
- **Leveraging External Expertise and Partnerships:** Another common strategy is that SMEs compensate for their lack of in-house expertise by bringing in external help. This could involve hiring IT consultants for specific projects, using government-funded SME tech advisory services, or partnering with tech startups/solutions providers. For example, an SME might work closely with a software vendor that provides not just a product but also training and customization for that SME's needs. Some join industry consortiums or local business networks to share knowledge about digital best practices. The notion of open innovation is visible, where SMEs collaborate or learn from others rather than innovate entirely internally. By leaning on external partners, SMEs can execute more

sophisticated digital strategies than they could alone. Of course, this requires identifying reliable partners, which is part of the SME's strategic acumen.

- **Continuous Learning and Adaptation:** A recurring theme is that SMEs which view digital transformation as a continuous journey rather than a one-time project tend to fare better ³⁰. These firms cultivate an attitude of ongoing learning – keeping up with new technological trends, continuously training their staff, and being willing to adapt strategies as they learn what works or as the environment changes. For instance, after implementing a new system, they actively solicit feedback, measure results, and refine their processes. If a particular digital tool isn't delivering benefits, they are willing to pivot to another solution. This learning orientation is essential given the fast pace of tech change. Sagala & Öri (2024) explicitly recommend that SMEs “*commit to investing in education and continuous learning*” as part of their digital strategy ³⁰. Anecdotally, SMEs that have successfully transformed often describe a multi-year evolution with several adjustments along the way, guided by an overarching commitment to become more digital.

Real-world examples from the literature illustrate these strategies. For example, a small retail SME in Italy might start by launching an online store (customer-facing priority), use a SaaS platform to do so (pragmatic off-the-shelf solution), and integrate it with their existing inventory spreadsheet. They might take it step by step: first getting basic e-commerce running, then adding analytics to track customer behavior, and later introducing AI-based recommendations. Throughout, the owner may seek advice from an e-commerce consultant or participate in a government digital training program (external support), and as sales grow, invest part of the revenue into further digital tools (continuous adaptation). Another case might be a manufacturing SME in Germany that begins with digitizing its production scheduling using a cloud ERP module, then gradually adds IoT sensors on critical machines (incremental approach). They align these tech adoptions with a goal of reducing downtime by, say, 20% (business-aligned objective). They involve their technicians in learning how to use IoT dashboards (workforce training) and perhaps collaborate with a university on implementing a predictive maintenance algorithm (partnership). Over a couple of years, they transform their operations significantly, but each step was measured and built on previous successes.

It is important to note that not all SMEs deploy these strategies effectively. Many SMEs still do not have a formal strategy and act opportunistically. The literature, however, suggests that the patterns above are associated with better outcomes. In particular, SMEs that plan and take incremental, aligned steps – and invest in their people – tend to navigate the digital transformation journey more successfully ⁴⁵. In contrast, those that either try to do too much at once without preparation, or do nothing until forced, often struggle.

Impact on Performance: Productivity, Innovation, and Customer Engagement

One of the critical questions for SMEs is whether digital transformation truly pays off in terms of business performance. Recent evidence in the literature indicates that, when executed well, digital transformation can have substantial positive impacts on several performance dimensions of SMEs:

- **Productivity and Efficiency Gains:** Numerous studies link the adoption of digital technologies with improved productivity in SMEs. Digitally mature SMEs often streamline their operations, automate repetitive tasks, and reduce errors, leading to higher output per unit of input. For example, implementing digital workflow tools or IoT-based monitoring can cut down the time and cost of operations. Quantitative assessments have found that even partial adoption of advanced

technologies can yield significant efficiency improvements. In the manufacturing sector, small firms implementing **Industry 4.0** technologies (like sensors, data analytics, robotics) experienced productivity increases ranging from single-digit improvements to as much as 50% in some cases ⁴⁶. One analysis cited by ITIF showed SME manufacturers boosting productivity by 7% to 26%, and another case up to 50%, after embracing suites of digital tools ⁴⁶. These are substantial gains that can translate into better competitiveness and profitability. Outside of manufacturing, service SMEs using digital platforms (for instance, an SME using an online reservation and payment system) also report efficiency gains, as processes that used to be manual or slow become automated and faster.

- **Innovation and Agility:** Digital transformation often expands an SME's capacity for innovation. By utilizing digital technologies, SMEs can develop new products or services (e.g., a traditional retailer creating an online consulting service, or a manufacturer offering a data-driven service on top of a physical product). They can also innovate in business models – such as moving from purely offline sales to a hybrid subscription model using digital channels. Research indicates that digitally active SMEs tend to introduce more product or process innovations than their less digital counterparts ⁶. The reasoning is that using digital tools exposes firms to new ideas and data, facilitates experimentation at lower cost, and often requires rethinking old ways – all of which are conducive to innovation. Furthermore, SMEs with digital capabilities have shown greater agility in pivoting during rapidly changing conditions. A case in point is the COVID-19 pandemic: SMEs that had invested in digital infrastructure were quicker to adapt (by shifting sales online, for instance) and even came up with creative responses (such as local restaurants offering app-based delivery or classes going virtual) – showcasing innovation under pressure. Some empirical studies associate high digital tool usage with a higher rate of new product launches and a faster time-to-market for SMEs. In essence, digital transformation can foster a culture and toolkit for continuous innovation.
- **Financial Performance and Growth:** Ultimately, SMEs are interested in whether digital transformation boosts revenues, profits, or growth rates. The literature provides growing evidence of a positive correlation. A study of European SMEs found that adoption of AI was correlated with significant revenue growth – *“positively affects SMEs’ revenue growth (>30%)”* according to the study's results ²⁵. Similarly, an analysis of Irish and UK SMEs using digital technologies showed higher sales growth and productivity compared to those that did not adopt such tools (controlling for other factors). Digital firms tend to reach wider markets (geographical expansion through online channels can drive revenue), improve customer retention (through better service), and optimize costs (through efficiency), all contributing to better financial metrics. Of course, the impact can vary: not every digital investment yields immediate profit, and some SMEs may overspend on technology without a clear return. But on average, the trend is that thoughtfully implemented digital transformation has a positive ROI. One concrete example comes from a Chinese manufacturing SME case study: by adopting a shared digital platform for production capacity (a form of digital transformation in the *“sharing economy”* context), the company significantly improved its profitability. Over a few years post-transformation, the SME's **main business profit margin** increased from about 32.6% to 45.9%, and its asset turnover ratio rose from 0.64 to 0.84, indicating more efficient use of assets – gains confirmed by statistical analysis ⁴⁷. This case underscores how integrating digital solutions (in this case, an external platform) can directly enhance financial performance in an SME setting.
- **Customer Engagement and Satisfaction:** Digital transformation often has a direct impact on how SMEs interact with customers – and by extension, on customer satisfaction and engagement.

Through digital channels, SMEs can communicate with and service customers faster and more effectively. For instance, an SME that implements a customer relationship management (CRM) system and data analytics might gain insights to personalize its offerings or marketing, leading to increased customer satisfaction. Social media engagement allows closer relationships and brand building. E-commerce and mobile apps provide convenience that modern customers expect. The literature notes that SMEs using digital tools can “*boost customer satisfaction through personalized products and services*” and even strengthen customer loyalty by offering a better experience ⁴⁸. In one study, introducing digital channels in small retail businesses improved customer retention rates, as customers appreciated features like online support and quick response times. Moreover, digital presence often increases an SME’s visibility, attracting new customers who primarily search and shop online. In summary, by meeting customers in the digital space with the right tools (ranging from simple social media interaction to AI-driven recommendations), SMEs can enhance engagement, which in turn drives repeat business and word-of-mouth referrals – key factors for growth.

- **Operational Resilience:** Though not always formally measured as a performance metric, an important benefit observed is improved resilience. SMEs that digitize some of their operations can better withstand disruptions. For example, a company with cloud-based data storage can recover faster from a local disaster than one with all data on a single office computer. Similarly, SMEs that developed omni-channel sales (both physical and online) were more resilient during lockdowns or market shifts. This resilience contributes indirectly to performance by reducing downtime and protecting revenue streams under adverse conditions. Policymakers and institutions, especially after COVID-19, have highlighted digital transformation as a means for SMEs to build resilience against future shocks ¹.

That said, the impact on performance is not automatic. There are cases where SMEs invest in technology but do not see immediate gains, often due to incomplete implementation or not aligning tech use with business strategy (e.g., installing software that employees don’t fully utilize). The literature warns that to realize performance gains, digital transformation must be accompanied by complementary changes in the organization, such as training employees and redesigning processes ¹⁸. When those pieces are in place, the evidence is increasingly clear that SMEs benefit substantially from going digital. One broad European survey concluded that “*digital firms tend to perform better than non-digital firms...they invest more, are more innovative, have better management practices, grow faster and create higher-paying jobs.*” ⁶. This encapsulates the multi-faceted performance improvements linked to digital transformation.

In summary, effective digital transformation can help SMEs do more with less (higher productivity), create new value (innovation), improve financial outcomes (growth in revenue and profit), and forge stronger relationships with customers (engagement and loyalty). These improvements allow SMEs not only to survive in a competitive environment but to potentially thrive and scale. The magnitude of benefits can vary by industry and the SME’s initial conditions – for example, a tech startup SME might see digital tools as core to its product offering, while a traditional small manufacturer might primarily gain efficiency – but virtually all SMEs have areas where digitalization can drive performance positively. The challenge is ensuring that SMEs approach transformation in a way that these benefits are realized, which ties back to having the right strategy and overcoming barriers. The next section (Discussion) will delve into what these findings mean for theory and practice, and what gaps remain.

Discussion

In reviewing the strategies and challenges of SME digital transformation, several important implications emerge for both theory and practice. Additionally, it becomes clear that while our understanding has advanced in recent years, there remain notable gaps in the research that present opportunities for future study.

Theoretical and Practical Implications

From a theoretical perspective, the findings reinforce and extend existing innovation and management theories, particularly in the context of smaller firms. One relevant theoretical lens is the **dynamic capabilities** framework, which posits that organizations need to continuously build, integrate, and reconfigure internal and external competences to address rapidly changing environments (Teece, 2007). Our review suggests that SMEs successful in digital transformation are essentially exercising dynamic capabilities: they *sense* the need for change (awareness of digital trends and opportunities), they *seize* opportunities by formulating strategies to adopt new technologies, and they *transform* their organization by realigning resources and processes to support the change ⁴⁹. For instance, developing digital awareness and literacy in the organization can be seen as building a sensing capability (recognizing where digital tools can help) ³¹. Crafting a digital strategy aligned with business goals reflects a seizing capability – leveraging opportunities through new processes or services ²⁹. Finally, the ability of an SME to reconfigure itself (e.g., altering workflows, training staff for new roles, perhaps dropping old product lines to focus on digital-enabled ones) corresponds to the transforming aspect. The literature's emphasis on continuous learning and incremental adaptation in SMEs aligns well with dynamic capability theory, highlighting that SMEs must constantly evolve their knowledge base and routines to keep up with technological change ³⁰. Thus, our findings provide empirical support that dynamic capabilities are critical at the SME level for digital transformation success.

Another theoretical implication concerns the **resource-based view (RBV)** and its extension, the resource-constrained perspective of SMEs. RBV suggests that firm performance differences stem from unique resources and capabilities. In SMEs, scarce resources (financial, human, technological) are a defining characteristic. The strategies we observed (incrementalism, leveraging external resources, etc.) illustrate how SMEs attempt to overcome resource limitations. This resonates with the concept of “*bricolage*” in entrepreneurship theory – making do with what is at hand. SMEs often creatively recombine available resources for digital initiatives (using existing staff to champion IT projects, repurposing off-the-shelf tech for their niche use). Our findings underscore that while limited resources constrain SMEs (supporting RBV's idea that lack of certain resources is a competitive disadvantage), SMEs can strategically compensate through agility and external resource mobilization ¹⁹ ³³. In effect, the agility of SMEs can be a resource in itself. This suggests a refinement to theory: in the digital era, an SME's intangible assets like adaptability, learning culture, and network connections may be as important as tangible assets in determining digital transformation outcomes. Traditional RBV might undervalue these dynamic, networked resources, so our review points to their importance.

Moreover, our comparative look highlights the role of the **institutional environment** – an angle of interest in institutional theory. Differences between regions (EU, USA, Asia) in SME digitalization reflect institutional support and culture. For example, German SMEs benefit from strong institutional backing (e.g., Industrie 4.0 programs, vocational training systems), whereas US SMEs operate in a more laissez-faire environment with less direct support, and Asian contexts like Japan or South Korea have their own unique policy

interventions. The observation that “Germany invests 6.2 times more (per GDP) than the US in SME digital modernization, and Japan 54 times more” ⁵⁰ indicates that institutional investment can be a differentiator in SME transformation success. This implies that theories of technology adoption must consider not only firm-internal factors but also external institutional factors like government initiatives, availability of infrastructure, and cultural attitudes toward technology. It aligns with the **Technology-Organization-Environment (TOE)** framework, which posits that adoption is influenced by organizational factors and the external environment. Our findings give specific weight to the “Environment” aspect: SMEs in supportive environments (access to funding, training, infrastructure) have a head start. The practical implication is clear – policymakers and industry bodies play a critical role in enabling SME digital transformation, corroborating arguments made in policy circles that a supportive ecosystem is needed ³⁹ ³³ .

From a practical standpoint, the review offers several lessons and recommendations for SME owners/managers and for policymakers or support organizations:

- **For SME Management:** One key takeaway is that having a clear strategy (even if modest) is crucial. SME leaders should take time to assess their business and craft a digitalization plan that aligns with their goals and capacity. This plan should prioritize actions (rather than attempting too much at once) and set realistic milestones. The evidence suggests that **incremental wins** build momentum – a small successful IT project can justify the next one. Therefore, SME managers might start with a project that is likely to show quick benefits (for example, implementing an online ordering system that could increase sales relatively quickly). Another practical implication is the importance of investing in **human capital**. SME managers should consider training programs for their staff or themselves, many of which are now available online or via government schemes, to build digital skills ³¹ ³² . Even basic digital literacy and awareness across the team can reduce resistance to new tools. Additionally, SMEs should not hesitate to **seek external help** – whether through hiring consultants for critical phases, partnering with tech firms, or joining peer networks. The cost of consulting or participating in a digital innovation hub can be outweighed by the time and errors saved. Practically, SMEs might leverage free or subsidized resources first (many countries have free digital advisory services for SMEs or public webinars on digital skills). Also, the role of **leadership** came out strongly: SME owners should lead by example in embracing technology. If the owner uses data in decisions and shows enthusiasm for new solutions, it signals to employees that digital transformation is a priority, helping shape a pro-change culture.
- **For Policymakers and Support Institutions:** The findings highlight several areas where external support can make a difference. **Financial incentives** (grants, tax credits, low-interest loans for digital adoption) can alleviate the resource barrier. Given that cost is a top hurdle, schemes that reduce the financial burden of, say, acquiring an ERP system or funding an IT hire for an SME, will encourage more firms to take the leap ³⁵ . **Training and education programs** are equally vital – governments and industry associations can provide training workshops, digital toolkits, and even one-on-one mentoring for SMEs. For example, the EU and various member states have initiated digital innovation hubs and “SME digitalization” programs which could be expanded and made easily accessible. Our review points to the importance of reaching the smallest firms (micro-enterprises) who might not normally engage with such programs. Ensuring awareness of available support is a practical challenge; outreach efforts may be needed (perhaps via local chambers of commerce or through partnerships with banks that interface with SMEs). Another implication is the need to develop **infrastructure**: SMEs depend on reliable internet, cybersecurity frameworks, and digital platforms. Policy can help by investing in rural broadband, creating trusted e-government platforms

(for digital IDs, signatures, etc., that facilitate business online operations), and setting up standards or guidance on cybersecurity tailored for SMEs. The fact that SMEs worry about data security and compliance suggests a niche for support – e.g., providing clear guidelines and affordable solutions for SME cybersecurity and data protection can remove some hesitation to digitalize ³⁷. Finally, fostering **networks and clusters** where SMEs can learn from each other or collaborate in digital projects is beneficial. For example, regional innovation clusters that include tech startups, SMEs, universities, and large companies can create an environment of knowledge sharing. The practical evidence shows that when SMEs are embedded in such networks, they have more exposure to digital ideas and partners that can help implement them ³³.

The above implications emphasize that SME digital transformation is a socio-technical process: it involves technology, people, and institutions. Theoretically, it calls for interdisciplinary understanding (combining insights from information systems, entrepreneurship, organizational behavior, and policy studies). Practically, it calls for a coordinated approach among various stakeholders to address the multifaceted barriers.

Gaps in Current Research

While research on SME digital transformation has grown rapidly in the past five years, our systematic review reveals several gaps and areas that warrant further investigation:

- **Quantitative Evidence on Performance Outcomes:** Although many studies qualitatively assert performance benefits of digitalization for SMEs, there is still a need for more rigorous quantitative evidence, especially linking digital transformation to financial performance over time. Few studies incorporate hard financial metrics (revenue growth, profit margins, productivity ratios) directly and longitudinally. As noted, some research rarely includes detailed financial analysis when examining SME digital initiatives ³⁴. Future research could use longitudinal data (e.g., panel data of firms) to assess how digital adoption impacts SME survival, growth, and profitability over multiple years. This would address the gap of causality: does digital transformation drive performance, or are better-performing SMEs simply more likely to invest in digital tech? More sophisticated econometric studies or quasi-experiments (for example, examining firms before and after receiving a digitalization grant) could strengthen the evidence base.
- **Research on Micro-Enterprises and Sectoral Differences:** Many SME studies focus on small and medium firms collectively, or even only on the upper range of SMEs. Micro-enterprises (e.g., businesses with <10 employees) are less frequently studied yet make up a huge portion of all firms. Their digital transformation challenges might be different (they often have zero IT staff, extremely limited budgets, and perhaps less formal structures). Likewise, sector-specific nuances are under-researched. For instance, digital transformation in manufacturing SMEs (Industry 4.0 adoption) has been studied somewhat, but what about small restaurants, craft producers, or local service businesses (like small accounting firms or healthcare clinics)? Each sector might have unique drivers and barriers. There is a gap in understanding how context-specific factors play out. Future work could delve into comparative studies across sectors, or target under-represented sectors to see if general findings hold.
- **Geographical and Cross-Cultural Studies:** Our focus was largely on Europe with comparisons to the US and Asia, but many regions (Africa, Latin America, other parts of Asia) have different contexts

for SMEs. The OECD and World Bank often highlight that SMEs in developing countries face additional challenges like weaker infrastructure and larger informal sectors. However, academic literature with empirical data from these regions is still limited (though growing). Sagala et al. (2024) noted that the phenomenon of SME digital transformation in developing countries remains understudied ⁵¹. Future research should explore these contexts – for example, how do SMEs in emerging economies leapfrog with mobile technology, or how do they use digital platforms to access global markets despite local constraints? Additionally, cultural factors (like attitudes towards risk, innovation, or hierarchical decision-making) can influence transformation; comparative studies that factor in cultural dimensions could fill this gap.

- **Depth on Organizational Change and Human Factors:** While technology adoption is documented, there is less research on the *organizational change process* within SMEs during digital transformation. We know skills and culture are important, but detailed case studies or ethnographies that follow SME employees and managers through a transformation could provide richer insights on how roles, identities, and power dynamics shift (or resist shifting) in a small firm. The family firm study we cited begins to touch on such issues (e.g., how overcentralization by an owner can impede change) ⁵²
³⁸. More research in that vein – possibly qualitative – is needed to understand how to effectively manage change in SMEs at the human level. For instance, what training methods work best for upskilling SME employees? How can small businesses cultivate a digital culture, especially if the workforce is older or less tech-inclined? These micro-level questions are crucial for practical implementation but not fully answered by current research.
- **Emerging Technologies and SMEs:** As technology evolves, new tools like blockchain, advanced robotics, or even quantum computing could have SME applications. Currently, topics like AI and IoT are just starting to be studied in SMEs (with AI adoption being low but rising). In a few years, AI might be more commonplace, and questions will shift to: how do SMEs govern AI usage, how do they address ethical and privacy issues, etc. The research hasn't caught up with those questions yet. Similarly, **sustainability and digital transformation** is an area gaining interest – how can SMEs use digital tech to also achieve environmental sustainability goals? For example, using data to optimize energy use or reduce waste. Integrating the sustainability angle is a gap in many studies that focus purely on economic performance. As the business world places more emphasis on sustainable practices, exploring the intersection of green goals and digital goals for SMEs would be timely.
- **Effectiveness of Policy Interventions:** We have various programs worldwide aiming to help SME digitalization (from training schemes to subsidies). Surprisingly, there's a lack of rigorous evaluation of these initiatives in academic literature. Do digital grants significantly accelerate SME transformation? What types of training yield the best outcomes? These are questions policymakers would love to have answers to, but scholarly research on them is limited. Collaborations between researchers and governments to analyze program data could fill this gap. For example, a study might compare SMEs that participated in a certain digital support program with a control group that did not, to measure differences in adoption or performance – providing evidence of what works. This kind of applied research would directly inform policy design.
- **Strategic Management in SMEs' Digital Era:** As noted by some authors, there is still “scant research on strategic considerations” for SME digital transformation ⁵³. This gap implies that frameworks and tools specifically to help SMEs strategize (not just adopt tech tactically) are underdeveloped. Future research could develop SME-specific strategic frameworks or maturity models for digital

transformation (some exist but often adapt large firm models). For instance, how should an SME prioritize digital investments? How to measure their digital maturity in a simple way? Addressing these could aid both academic understanding and give practical planning tools for SMEs.

By identifying these gaps, we set the stage for the next section, which will outline future research directions. Addressing these gaps will deepen our understanding of SME digital transformation and help ensure that the knowledge keeps pace with the evolving digital landscape. In particular, filling these gaps will be vital to support the millions of small firms globally that are just beginning their digital journey or will do so in the coming years.

Conclusion & Future Research Directions

Digital transformation is no longer optional for SMEs – it has become a defining feature of competitiveness and resilience in the contemporary economy. This paper reviewed recent literature (2020–2025) on the strategies and challenges of digital transformation in SMEs, with a focus on European small businesses and comparative insights from other regions. We highlighted that SMEs, despite accounting for the majority of businesses and employment, lag behind larger firms in digital adoption due to financial, technological, and organizational barriers. At the same time, there is clear evidence that those SMEs which successfully embrace digital tools can reap significant benefits in productivity, innovation capacity, market reach, and customer satisfaction, ultimately leading to better performance and growth.

Key conclusions from our review include: (1) Effective SME digital transformation typically involves aligning technology adoption with the firm's specific context and strategy, using incremental and agile implementation to manage risk and resource limits ³⁰. (2) Critical success factors such as top management support, employee skill development, and openness to change must be in place to turn digital investments into tangible performance improvements ¹⁷ ²⁹. (3) Common barriers like lack of financing, skills deficits, cultural resistance, and infrastructural gaps are significant hurdles, but they can be mitigated through targeted actions both within the firm and via external support ³⁵ ³⁷. (4) Regional and institutional differences matter – SMEs in environments with strong digital ecosystems (training, incentives, infrastructure) have an advantage, suggesting a role for policy in leveling the playing field ⁵⁰. (5) Digital transformation is a continuous journey rather than a one-time project; SMEs must adopt a mindset of continuous learning and adaptation to sustain competitive advantage in a fast-changing digital landscape ³⁰.

Looking ahead, we propose several **future research directions** to address the gaps identified and to support SMEs in their digital endeavors:

- **Longitudinal and Impact Studies:** Future research should employ longitudinal designs to track SME digital transformation over time, quantifying its impact on performance. For example, studies could follow a cohort of SMEs as they implement digital projects and measure changes in productivity, revenue, or survival rates year by year. Causal inference methods (like difference-in-differences or randomized experiments if feasible) would greatly strengthen evidence on the ROI of digital transformation for SMEs.
- **Micro and Sector-Specific Analyses:** More fine-grained studies focusing on micro-enterprises and specific industries are needed. How a five-person craft business digitalizes may differ greatly from a 200-employee manufacturing SME. Researchers could examine case studies or survey data in under-

researched segments (e.g., micro-retailers, small farms using digital agriculture, local service providers adopting platforms) to broaden our understanding of the breadth of SME transformation experiences. Sector-specific digital transformation frameworks could be developed to provide tailored guidance (for instance, a “digital roadmap for small retailers” vs. one for small manufacturers).

- **Cross-Country Comparative Research:** There is value in expanding research to diverse geographies, especially emerging markets. Comparative studies could explore how SME digitalization unfolds under different economic and cultural conditions. For example, how do SMEs in Southeast Asia or Africa leverage mobile technologies versus those in Europe? What barriers are universal and which are context-dependent? Such research can inform international development efforts and show how lessons from one region might transfer (or need adaptation) to another. Collaborations between researchers across countries can help collect and analyze comparative data.
- **Human and Organizational Dynamics:** Future qualitative research should delve deeper into the human side of SME digital transformation. In-depth interviews, ethnographies, or process studies can shed light on how SME owners and employees perceive digital change, how roles and power structures shift, and how companies manage the psychological aspects of transformation. Special attention could be given to family-owned SMEs (which have unique dynamics) and to workforce development processes in small firms (e.g., what training methods most effectively upskill employees in digital competencies). Understanding these dynamics will help refine change management approaches specifically suited for SMEs.
- **Emerging Technologies & Innovation in SMEs:** As new technologies emerge, research should examine their adoption and impact on SMEs. For instance, as artificial intelligence tools become more accessible (think of no-code AI platforms), how are SMEs beginning to use them for marketing, customer service, or operations? Are there examples of small firms successfully implementing, say, blockchain for supply chain traceability or 3D printing for rapid prototyping? Studying early adopters of such technologies among SMEs can provide insights into opportunities and pitfalls, guiding others. Additionally, research on how digital transformation contributes to SMEs’ innovation ecosystems (for example, how SMEs collaborate with startups or participate in digital platforms to innovate) would be valuable.
- **Policy Evaluation and Best Practices:** Researchers should partner with governments and industry bodies to evaluate the effectiveness of current SME digitalization initiatives. Through surveys or interviews with participants in these programs, one can identify which support measures are most valued by SMEs and lead to actual improvements. Best practices can then be distilled – for example, what elements should a successful SME digital training program include, or how to structure a grant scheme to maximize uptake and impact. Sharing such findings can help policymakers worldwide refine their strategies to support SME digital growth. Furthermore, normative research could propose new policy ideas, such as novel public-private partnership models or community-based approaches to helping groups of SMEs transform together.
- **Digital Transformation and Sustainability Goals:** An emerging area is the intersection of digitalization and sustainable business practices. Future studies could explore how SMEs can use digital technologies not only for profit, but also for environmental and social objectives – for example, using IoT and analytics to reduce energy consumption, or e-commerce to enable circular

economy models for small producers. This aligns with the increasing interest in “twin transition” (digital and green) in Europe and elsewhere. SMEs have a role to play in this, and research can highlight synergies and trade-offs they face when pursuing digital sustainability initiatives.

In conclusion, digital transformation offers immense promise for empowering SMEs, allowing them to enhance efficiency, innovate, and compete in ways that were previously impossible on a small scale. Yet, it also presents non-trivial challenges that many SMEs are still grappling with. This systematic review has synthesized current knowledge on how SMEs can strategize for and overcome the hurdles of digitalization. The path forward will require collaborative efforts: SMEs taking initiative to adopt and learn, larger corporations and tech providers creating inclusive solutions for smaller clients, policymakers crafting enabling environments, and academics and educators providing insights and training. By addressing the identified gaps and following the outlined research directions, we can build a more comprehensive and nuanced understanding of SME digital transformation. Such knowledge will ultimately contribute to guiding millions of SMEs worldwide to successfully navigate their digital journeys, fostering inclusive economic growth and innovation in the digital age.

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